



Mitigating not-invented-here and not-sold-here problems: The role of corporate innovation hubs

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ABSTRACT

Despite a growing number of Corporate Innovation Hubs (CIHs) in recent years, limited attention has been paid to understanding the key problems that arise among organizations collaborating through CIHs. In particular, organizations often experience Not-Invented-Here (NIH) and Not-Sold-Here (NSH) problems, i.e. negative attitudes towards absorbing external knowledge and towards sharing internal knowledge externally. Consequently, many CIHs fail to deliver and are regarded as “innovation theatres” rather than engines of renewal. By drawing upon an inductive multiple case study of five CIHs, their parent companies and associated startups, located in Silicon Valley (USA) and the Gothenburg region (Sweden), the article sheds light on how CIHs can mitigate NIH and NSH problems in knowledge transfer. Specifically, we investigate the causes, consequences and mitigating mechanisms of NIH and NSH problems among the organizations collaborating through a CIH. These findings are presented in a framework that connects causes and consequences with the corresponding mitigating mechanisms. We also present new theoretical implications for the literatures on NIH and NSH.

1. Introduction

A Corporate Innovation Hub (CIH) mediates knowledge transactions and scouts for new ideas, concepts and technologies (Berger and Brem, 2016; Monteiro and Birkinshaw, 2017) by acting as an intermediate between incumbents and startups (Crisan et al., 2019). CIHs act as engines of renewal by creating bridges into innovative ecosystems where key organizations can collaborate to generate value by exchanging knowledge (Kupp et al., 2017; Walrave et al., 2018). For example, Honda established Honda Innovations where they have collaborated with over 40 startups and recently announced a partnership with SoundHound Inc., a leading innovator in voice-enabled AI (Fehrenbacher, 2019; Honda, 2018). The establishment of Honda Innovations enabled knowledge transfer among involved organizations in Silicon Valley to generate new business and innovative technologies (Ohnsman, 2017).

Knowledge transfer is defined as a process by which knowledge gained from one situation can be applied to another (Singley and

Anderson, 1989). CIHs, typically established by large incumbent companies such as Honda, Ford, IBM, Walmart or Microsoft (Berger & Brem, 2016; Bhan Ahuja, 2019; Hausberg & Korreck, 2018), are vital for knowledge transfer (Kötting, 2019). CIHs both absorb and share knowledge (Aquilani et al., 2017) among their collaborating organizations as they mediate knowledge transactions (Kohler, 2016; Zahra, 2015). The absorption of knowledge happens when transferring knowledge from external organizations (such as startups) into an incumbent company, whereas sharing of knowledge occurs when transferring corporate internal knowledge to external organizations. The organizations involved thus explore new ideas and opportunities together, and exchange different types of knowledge (Schuh et al., 2017).

However, achieving collaboration and knowledge transfer through CIHs is far from easy. One of the challenges is that the organizations involved are fundamentally different in terms of company culture and management style (Van de Vrande, De Jong & De Rochemont, 2009; Weiblen and Chesbrough, 2015). Many CIHs also report problems with

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opportunistic behavior and a lack of peer support when bringing ideas from a CIH to the parent company (Fecher et al., 2020; Longo and Giaccone, 2017; Monteiro, 2015).

For these reasons, CIHs need to deal with two key challenges in knowledge transfer: the “Not-Invented-Here” (NIH) and “Not-Sold-Here” (NSH) problems (Antons et al., 2017; Chesbrough, 2003; De Araújo Burcharth et al., 2014; Berger and Brem, 2016). Chesbrough (2003) describes NIH and NSH as two viruses, which create an allergic reaction when transferring knowledge among organizations.

The NIH problem is a negative attitude to external knowledge (Katz and Allen, 1982), resulting in a rejection of external ideas (Hjalmarsson, Jordanius, Juell-Skielse, & Kailas, 2019). NIH is a major problem in inbound open innovation processes (Chesbrough, 2003; Cohen and Levinthal, 1990). As Antons and Piller (2015, p.197) point out, NIH results in “... negatively shaped attitudes of an individual towards knowledge that has to cross a contextual disciplinary, spatial or organizational (functional) boundary, resulting in either suboptimal utilization or its rejection”.

In contrast, NSH is a protective attitude toward external knowledge exploitation (e.g. via joint-venture relationships, licensing agreements, or free revealing) (Bianchi et al., 2011; De Araújo Burcharth et al., 2014). NSH hinges on the fear of losing competitive advantage when transferring a firm’s internal knowledge assets to outsiders or sharing them with other organizations, and is regarded as a major problem in outbound open innovation processes (Chesbrough, 2011). NSH may result in mediocre solutions, terminated projects, or technology left sitting on the shelf (Grönlund et al., 2010). Arguably, both NIH and NSH hinder effective knowledge transfer among startups and parent companies collaborating through a CIH (Antons et al., 2017; De Araújo Burcharth et al., 2014). For example, Hannen et al. (2019) provide evidence from 565 global R&D projects that NIH is widely common and hinders the absorption of external knowledge. When one or both of these syndromes are present, organizations collaborating through a CIH will not be able to capitalize fully on their partnerships, and the CIH fails to achieve its mission.

We address two gaps in the extant literature on NIH and NSH. First, prior literature on NIH and NSH problems has a bias towards dyads, i.e. knowledge transfer between two companies or countries (Antons and Piller, 2015; Aquilani et al., 2017; De Araújo Burcharth et al., 2014; Hussinger and Wastyn, 2012; Kathoefer and Leker, 2012). Few prior studies thus consider the context of intermediaries such as CIHs to enable knowledge transfer among multiple organizations (Berger and Brem, 2016; Kohler, 2016) from an ecosystem perspective (Hannen et al., 2019), which is often the case in practice. For example, prior studies have mainly focused on knowledge transfer between a parent company and a CIH (Monteiro and Birkinshaw, 2017), thus excluding startups. And the limited number of studies that exist (e.g. Aquilani et al., 2017) conclude that both NIH and NSH problems arise in intermediaries exploring ideas and transferring knowledge, but these studies are based solely on conceptual analyses of the literature. Second, most prior research focuses on either NIH or NSH, or proposes generic remedies for both problems. For example, Schuh et al. (2017) highlight the importance of understanding NIH in knowledge transfer through a CIH but do not consider the importance of and Aquilani et al. (2017) propose trust to mitigate both problems. In open and coupled innovation processes, which are characterized by active inward and outward technology transfer, both NIH and NSH should be considered (Frishammar et al., 2019) and appropriate mitigation mechanisms may be different for the two problems. Specifically, prior research has demonstrated that CIHs are often struggling in knowledge transfer (Fecher et al., 2020; Hannen et al., 2019), but the literature is still largely tacit regarding the specific hindering factors in relation to NIH and NSH problems.

For CIHs to successfully act as intermediaries enabling knowledge

transfer among their collaborating organizations, especially their parent company and startups, more research is needed to understand the specific hindering attitudes and their mitigating mechanisms (Del Sarto, Marullo & Di Minin, 2018; Fecher et al., 2020; Hannen et al., 2019). In addition, increased awareness of the consequences is a step towards better understanding NIH and NSH problems and their causes. This article, therefore, sets out to enhance the literature of NIH and NSH problems in knowledge transfer, from the perspective of a CIH (Del Sarto et al., 2018; Schuh et al., 2017; Zahra, 2015). The overall purpose of our study is to ascertain the following: *How can corporate innovation hubs mitigate not-invented-here and not-sold-here problems in knowledge transfer?* To operationalize this purpose, the following three research questions are addressed: *What are the causes of NIH & NSH problems among organizations collaborating through a CIH? What are the consequences of NIH & NSH problems among organizations collaborating through a CIH? Finally, What mechanisms can a CIH use to mitigate NIH & NSH problems among collaborating organizations?*

2. Theoretical background

2.1. CIHs and knowledge transfer

CIHs are typically situated in clusters of connected companies and institutions based on technological or economic concentrations (Porter, 1998). In these clusters, innovation ecosystems evolve, of which CIHs can form an integral part (Adner, 2017). Other key organizations are startups, incumbent corporations, investors, and academic institutions, all of which may collaborate when their interests align (Adner, 2017; Salomaa, 2018). In other words, a CIH is part of a structured micro-ecosystem where organizations seek to explore potential value propositions together by fostering the dynamics of innovation and entrepreneurship (Adner, 2017; Banc and Messegheem, 2020). However, not all CIHs are the same (Pauwels et al., 2016; Weiblen and Chesbrough, 2015), yet they share the mission of generating new ideas, concepts and technologies to bring to their parent company. Some CIHs are more focused on technology scouting and exploration, some engage in academic research, and still others focus on generating new business solutions together with partners. CIHs are thus, in some respects, similar to corporate incubators and accelerators. Incubators and accelerators are, however, broader and more all-encompassing phenomena. Accelerators can be considered to be a particular form of incubators; this is discussed in detail by e.g. Hallen et al. (2020) and Hausberg and Korreck (2018). CIHs differ from corporate accelerators, which often focus on scaling startups (Kanbach and Stubner, 2016; Monteiro and Birkinshaw, 2017). In this study we view CIHs as independent and boundary-spanning corporate units set up outside ordinary business structures which mediate knowledge transactions and whose mission is to scout for new ideas, concepts and technologies (Berger and Brem, 2016; Monteiro and Birkinshaw, 2017).

The purpose of a CIH is thus to explore new ideas, technologies and knowledge that can be used to generate new customer solutions or new business by performing an intermediation role (Euchner, 2016; Kohler, 2016). A CIH thus plays a key role in transferring knowledge among its parent company and associated startups, where the knowledge transferred can take different forms, such as domain-specific knowledge in the form of technical expertise, datasets, patents, etc., or procedural knowledge in the form of product development methodologies or guidelines (Schuh et al., 2017). CIHs are thus different from independent intermediaries in open innovation. Independent intermediaries are stand-alone entities seeking to mediate between organizations seeking external solutions for innovation-related problems and potential solution providers (Diener et al., 2020). By contrast, the key objective of a CIH is to supply the parent corporation with whatever knowledge is

needed, but knowledge transfer is often two-way, in a reciprocal process. That is to say, parent corporations can also supply startups with critical knowledge through CIHs (Kohler, 2016).

2.2. Causes and consequences of NIH & NSH problems

On an overall level, Katz and Allen (1982) argued that the presence of NIH will generate incorrect evaluations of external ideas, leading to difficulties in seeing possible benefits and a decline in communication with external organizations. Prior literature on NIH and NSH problems among organizations collaborating through CIHs is limited, but an intermediate hypothesis is that CIHs and their collaborating organizations experience significant problems in knowledge transfer. For example, Chesbrough (2003) argues that NIH and NSH are counterpart syndromes, both hampering and undercutting the effectiveness of knowledge transfer among organizations when collaborating in an open innovation setting. The author further argues that NIH is more likely to be present in R&D projects; while NSH is more often found in projects centering on the business side.

Prior literature traces NIH problems to multiple causes. First, Antons and Piller (2015), in an extensive literature review of NIH, argue that NIH derives from psychological ownership, selective information processing, confirmation bias and disciplinary background. Psychological ownership occurs when specialists have a desire to defend their expert status, which can be at odds with new knowledge that is put into use. Selective information processing is an issue when managers experience difficulties in making evaluations when they and others involved approach a situation from different backgrounds and disciplines, and are thus led to contradictory conclusions. In contrast, confirmation biases are triggered by incentives and prior knowledge and understandings, e. g. when peoples' actions are directed towards the best outcome for themselves. Lastly, differences in individuals' disciplinary backgrounds may imply that startups and mature corporations within certain domains might be more affected by NIH than those in other domains. However, while the review by Antons and Piller (2015) provides important insights, it was not carried out in the context of CIHs. Second, Kathoefer and Leker (2012) propose that the likeliness of NIH depends more on personal traits than disciplinary background. This study was also, however, outside the context of CIHs. Third, Chesbrough (2006) argues that NIH is due to the fear of failure in selecting the right external ideas to work with. Fourth, Hussinger and Wastyn (2012) argue that NIH is more likely to occur when sourcing knowledge from competitors. Both these latter studies provide important insights into the NIH phenomenon, but also disregard the CIH context. For example, the ways that CIHs, startups and parent companies perceive each other's knowledge will greatly affect NIH problems, but arguably also its potential solutions. By taking an ecosystem perspective (Hannen et al., 2019), new insights into causes and consequences may follow (and by extension also new ideas for mitigating mechanisms). If these causes are not properly managed, NIH results in individuals underestimating the value of external knowledge and overestimating the efforts needed to absorb it, leading to either rejection of new knowledge or poorly performed implementations of such knowledge, and hence suboptimal decision-making (Antons and Piller, 2015).

In contrast, prior literature traces NSH problems to three causes. First, De Araújo Bianchi et al. (2011) argue that the protective attitudes at the core of NSH problems are driven by social influences embedded within a corporation's innovation culture, including norms, values and principles. As described by the authors, "... employees feel that if the knowledge or technology cannot be exploited in own products or markets, it should not be exploited at all" (p. 151). Second, Aquilani et al. (2017) propose that the NSH problem may be linked back to a lack of openness, where companies face the dilemma of wanting to share knowledge externally but are stuck in old internal patterns of innovation. Likewise, Hoegl et al. (2011) argue that NSH attitudes may be strengthened by limited prior experience with outward technology transfer and the

inefficiencies in markets for technological knowledge. Many companies may therefore have a stronger emphasis on exploiting internal knowledge. Third, Kline (2003) argues that NSH tendencies manifest through the fear of undermining a company's ability to retain fair value for its intellectual property rights or internal disagreements among employees about whether to license a technology or not. In a similar vein, Herzog (2011) argues that NSH tendencies can manifest through the fear of ending up in a legal dispute because employees are uncertain as to what knowledge they are allowed to share. It should be noted that none of these studies is in the context of CIHs, although Aquilani et al. (2017) theorize on intermediaries—theirs is, however, a purely conceptual paper.

If the causes of NSH problems are left unmanaged, negative NSH attitudes may result in fruitless knowledge exploitation, where startups and a parent corporation engage in collaboration without clearly showing their abilities, resulting in a situation where they are unable to make mutually beneficial exchanges. This situation might lead to frustration among employees at both startups and parent companies as their ideas, knowledge and technology are kept hidden and withheld from future development (Chesbrough, 2006). Arguably, this could lead to a feeling of distrust and disrespect (Kline 2003), resulting in demotivated employees who might even resign. As a result, companies with significant NSH tendencies often fail to profit from outbound open innovation (Hoegl et al. (2011).

2.3. Mitigating NIH & NSH problems in knowledge transfer through CIHs

Aquilani et al. (2017) argue that intermediaries engaging in knowledge transfer among organizations, in a similar manner to a CIH, can mitigate NIH and NSH through a strong sense of trust and openness. This is further supported by Longo and Giaccone (2017), who argue that non-monetary awards and informal control mechanisms, while encouraging employees at CIHs to be proactive and committed towards common goals, mitigate conflicting personal interests. However, and specifically for NIH, Aquilani et al. (2017) argue that it can be managed by establishing new roles, such as an idea scout, idea connector, integration expert, technological gatekeeper or champion of innovation. The authors also propose that NIH problems can be eliminated altogether by choosing the right ideas and then fostering these ideas and providing appropriate contacts while integrating the ideas into internal R&D processes. In contrast, to manage NSH, companies need to identify the right partners to collaborate with, and be able to distinguish the sale of IP rights from startups or spin-off creations (Aquilani et al., 2017).

Summing up, research into the causes and consequences of NIH and NSH problems in a CIH context is scarce, and for potential remedies (i.e. mitigation mechanisms) it is even scarcer. Most of the existing so-called mitigating mechanisms are thought to mitigate both problems at the same time, without any deeper thought about each problem's specific root causes. The advice provided by prior literature is also quite generic. For example, Aquilani et al. (2017) argue that a strong sense of trust is needed by CIHs to mitigate NIH and NSH problems, but do not specify how to accomplish this. Second, both Aquilani et al. (2017) and Longo and Giaccone (2017)—the two studies that focus directly on CIHs—lack sufficient empirical data to confirm proposed ideas. In particular, there is almost no prior literature on NIH and NSH problems in knowledge transfer through CIHs among collaborating organizations. Given the state of prior literature, we opted for an inductive multiple case study to better understand NIH and NSH problems in a CIH context. The next section describes our methods for doing so in detail.

3. Method

3.1. Research strategy and case selection

Despite the research conducted over the past decades, the literature on NSH is still nascent. As prior research on knowledge transfer through

CIHs with multiple actors involved is also nascent, we adopted an inductive method (Gioia et al., 2013; Strauss and Corbin, 1998). The inductive approach allowed us to build theory and to let our understanding of the research area progressively emerge, as the use of initial exploratory interviews gave the empirical study its direction and provided the theoretical background with a starting-point. The Gioia method has emerged as the most popular one for inductive case analysis in leading journals (Del Sarto et al., 2018; Foege et al. 2019; Leone 2020; Mees-Buss et al., 2020; Sjödin et al. 2019). Recently published work also shows that this method is applicable to analyzing research questions which focus on answering “what” (Shankar and Clausen, 2020; Zhao et al., 2020).

To fulfill the purpose, multiple case studies of knowledge transfer among startups and parent companies collaborating through CIHs were conducted. Needless to say, a CIH is made up of individuals doing the work, so the individual is the main unit of observation. But their views have been aggregated to shed light on the CIH, since it is the CIHs that constitute the main unit of analysis.

We investigated five CIHs, each of which acts as an intermediate in sharing knowledge between parent companies and startups in its surrounding ecosystem. The cases were selected on three criteria. First, each CIH was operating in the automotive industry; an industry in rapid technological transformation under the ACES trends—autonomous driving, connected cars, electrified vehicles, and smart mobility. The automotive industry plays an important role in the global economy and for GDP growth (Saber, 2018). Since 2010, investors have invested \$220 billion into more than 1100 companies across ten technology clusters in this area (Holland-Letz et al., 2019). While the automotive industry overall is heterogenous, the parts of it to which the CIHs belong were homogenous. Second, each CIH engaged with startups and was owned by its corresponding parent company, which created a dependency of knowledge transfer among all the organizations in the

ecosystem around the CIH. All CIHs were connected to a corporate venture capitalist (CVC) at the parent company to assure that promising new ideas, concepts or ventures could be financed. It should be noted that the number of CVC investments from Nordic corporates has more than doubled over the past five years (Oxford Research, 2019). Third, all CIHs actively explored novel transportation technologies through external collaborations, an area where NIH and NSH tendencies are high due to a tradition of developing technology in-house under a “closed innovation” logic (Rise Research Institutes of Sweden, 2018). For instance, CIH I is collaborating with startup Epsilon on automated driving systems at their location in Silicon Valley. This collaboration involves engineers from the parent company in Sweden, in an open innovation setting where NIH and NSH problems occur in this two-way reciprocal process of knowledge transfer.

All CIHs are briefly presented in Table 1. Since a few of the CIHs are hosted by larger R&D centers, the dedicated numbers of employees at each CIH unit is presented to facilitate comparability. It is important to point out that a larger CIH was able to execute more projects and activities than a smaller one, but the actual activities they performed were similar. All the CIHs studied are located in Silicon Valley, a global hub for high-tech innovation over the past 20 years (Cellan-Jones, 2020), but their parent companies are based in Sweden or Germany. The innovation ecosystem of the San Francisco Bay Area is one of the most venture capital heavy areas in the world. In 2018, 99.5 billion US dollars' worth of VC investments were made in the US, and 46% (45.6 billion US dollars) of those can be traced to Silicon Valley (PwC & CBInsights, 2019). At the same time, both Germany and Sweden score high on global innovation index rankings, where Sweden comes in second place after Switzerland (Cornell, 2019). Regarding the automotive industry, Gothenburg is a growing region receiving multiple international strategic projects and initiatives (Hermansson, n.d.).

Table 1
Key Information about the CIHs studied.

Corporate Innovation Hub	Description of Parent Company	CIH Location	Headquarter Location
CIH I Their objective is to leverage external innovation within digital technology through testing and developing new customer solutions together with organizations such as startups and tech corporations. Enable knowledge transfer in joint projects with the parent company. Established: autumn 2018. Number of employees at CIH: 5–10. Number of employees dedicated to CIH activities: 5–10.	Alfa European Automotive OEM, produces and distributes trucks, buses, construction equipment, as well as marine and industrial drive systems	United States	Sweden
CIH II Embrace new technologies, market trends and build capabilities within electrification and automation technology for future solutions. This is accomplished through collaborations with organizations such as startups, students and corporations, as well as connecting the parent company to joint projects. Established: autumn 2016. Number of employees at CIH: 100–200. Number of employees dedicated to CIH activities: 10–15.	Beta European Automotive OEM, produces and distributes cars	United States	Sweden
CIH III Sources world-class knowledge and talent for the future. This is accomplished by establishing collaborations with various organizations and by scouting for new technological advances. Established: 2018. Number of employees at CIH: 5–10. Number of employees dedicated to CIH activities: 5–10.	Gamma European Software company, develops and distributes software for driving systems	United States	Sweden
CIH IV Fosters innovation, collaboration and creativity within technology areas such as sensors, connectivity and automation. This is accomplished by interacting and collaborating with organizations such as startups, universities etc. Established: 1999. Number of employees at CIH: 200–250. Number of employees dedicated to CIH activities: 15–20.	Delta European supplier of automotive hardware and software, such as powertrain and injection technology, as well as steering and driving systems	United States	Germany
CIH V Delivers mobility solutions fit for the next generation. This is done by co-creation with startups within technology areas such as mobility and connectivity. The CIH has so far hosted 22 startups. Established: 2017. Number of employees at CIH: 3–5. Number of employees dedicated to CIH activities: 3–5.	Alfa, Beta and Gamma European CIH shared among multiple corporations to collectively engage with startups.	Sweden	Sweden

3.2. Data collection

Primary data was collected through semi-structured interviews in three waves over five months, from January to May of 2019. A total of 39 interviews were conducted, with 31 unique respondents. Of these, 14 were from CIHs, 12 from parent companies, 9 from startups, 2 from a research institute, 1 from a co-working space and 1 from a venture capital firm. The duration varied between 15 and 94 min—see Table 2 for an overview of the sample. Each interview was recorded and then transcribed. Interviews held in English were transcribed verbatim, whereas those held in Swedish were first transcribed in Swedish and then translated into English for analysis.

The *first wave* of interviews was intended to increase our understanding of the problems occurring in knowledge transfer processes through CIHs. Selected respondents provided valuable complementary information which enhanced our understanding of the most central organizations in such collaborations. For example, the venture capitalist interviewed had extensive knowledge from working with several CIHs. This first wave consisted of two sub-phases; one conducted in Gothenburg, Sweden, and the second in Silicon Valley, US, which spanned from late January to mid-March of 2019. The interviews used open-ended questions (such as “What challenges occur in these collaborations when transferring knowledge?”). A total of 9 interviews were held in Sweden and another 10 in the US.

The *second wave* of interviews was intended to seek more in-depth answers to the research questions, which had now been specified more fully. All respondents were chosen either based on previous interviews

or recommendations from the first wave.

Respondents from startups were selected based on previous or ongoing collaborations with any of the CIHs being researched. Semi-structured interview questions were used, which enabled us to ask follow-up questions. These semi-structured questions were based on insights from the theoretical background as well as from the first wave of interviews. Questions such as “Why do you believe NIH/NSH problems occur in these projects?” and “How can such problems be mitigated by a CIH?” were asked. The interview questions were constantly edited and improved as wave two unfolded. In total, 18 interviews were recorded, transcribed and analyzed, and language differences were treated as in wave one. The second wave spanned the second half of March 2019.

The *third wave* of interviews was intended to confirm and consolidate the results, and to fill in remaining blank spaces. In this wave, we selected respondents that seemed to be the most representative of a CIH and its corresponding parent company. Two confirmative interviews were held, one at CIH I and one at parent company Alfa during May of 2019. Lastly, we relied on triangulation where data from the interviews and the emerging framework were discussed (Jick, 1979), i.e. to verify the different respondents’ viewpoints against each other. This was done through one workshop held in the US and two presentations with a subsequent discussion section which were held in the US and Sweden, respectively. The workshop was held with 6 respondents from both CIHs and Startups. The respondents were asked to react to, for example, how the causes, consequences, and mitigating mechanisms of NIH and NSH problems were labeled, and then were tasked with drawing connections among these. The workshop resulted in three alternative frameworks

Table 2
Overview of sample and respondents.

Respondent	Wave	Position	Organization	Country	Date	Duration	Type
R1	1	Project Manager	Research Institute Zeta	SWE	2019-01-29	51 Min	F2F
R2	1	Chief Technology Officer	Parent Company Alfa	SWE	2019-01-30	47 Min	F2F
R3	1	CEO/Co-Founder	Startup Theta	SWE	2019-01-30	16 Min	F2F
R4	1	Project Manager	Parent Company Alfa	SWE	2019-01-31	30 Min	F2F
R5	1	CEO/Founder	Startup Eta	SWE	2019-02-01	30 Min	VIDEO
R6	1	Director of CIH	CIH V	SWE	2019-02-04	40 Min	F2F
R7	1	Venture Capitalist	Parent Company Alfa	SWE	2019-02-05	41 Min	F2F
R8	1	Open Innovation Manager	Parent Company Alfa	SWE	2019-02-08	29 Min	F2F
R9	1	Project Manager	Research Institute Zeta	SWE	2019-02-12	46 Min	F2F
R10	1	Director of CIH	CIH I	US	2019-02-19	50 Min	F2F
R11	1	Director of CIH	CIH II	US	2019-02-19	77 Min	F2F
R12	1	Business Developer	CIH II	US	2019-02-21	59 Min	F2F
R13	1	Business Developer	CIH II	US	2019-02-22	63 Min	F2F
R14	1	Head of Sales and Strategy	Startup Omega	US	2019-02-27	24 Min	F2F
R16	1	Director of CIH	CIH III	US	2019-02-28	62 Min	F2F
R17	1	Innovation Advisor	Co-working Space Kappa	US	2019-03-01	45 Min	F2F
R18	1	CEO/Founder	Startup Sigma	US	2019-03-04	94 Min	F2F
R19	1	Senior Innovation Manager	Parent Company Alfa	US	2019-03-12	51 Min	F2F
R20	1	CVC Adviser	Venture Capital Psi	US	2019-03-13	45 Min	F2F
R21	2	CEO/Founder	Startup Epsilon	US	2019-03-14	40 Min	F2F
R2	2	Chief Technology Officer	Parent Company Alfa	SWE	2019-03-15	65 Min	VIDEO
R22	2	Venture Capitalist	CIH IV	US	2019-03-15	65 Min	F2F
R23	2	Venture Capitalist	CIH I	US	2019-03-18	44 Min	F2F
R24	2	Tech Scout	CIH IV	US	2019-03-19	59 Min	F2F
R7	2	Venture Capitalist	Parent Company Alfa	SWE	2019-03-20	68 Min	VIDEO
R10	2	Director of CIH	CIH I	US	2019-03-20	62 Min	F2F
R25	2	Business Developer	Parent Company Beta	SWE	2019-03-21	47 Min	VIDEO
R26	2	Chief Technology Officer	Parent Company Gamma	SWE	2019-03-21	40 Min	VIDEO
R12	2	Business Developer	CIH II	US	2019-03-21	44 Min	F2F
R5	2	CEO/Founder	Startup Eta	SWE	2019-03-22	48 Min	VIDEO
R27	2	Specialist	Parent Company Alfa	SWE	2019-03-22	55 Min	F2F
R28	2	Software Developer	CIH II	US	2019-03-22	35 Min	F2F
R14	2	Head of Sales & Strategy	Startup Omega	US	2019-03-22	55 Min	F2F
R29	2	CEO/Co-Founder	Startup Iota	US	2019-03-26	27 Min	VIDEO
R30	2	CEO/Co-Founder	Startup Pi	US	2019-03-26	68 Min	F2F
R31	2	Legal Advisor	Parent Company Alfa	SWE	2019-03-28	44 Min	VIDEO
R16	2	Director of CIH	CIH III	US	2019-03-29	41 Min	F2F
R10	3	Director of CIH	CIH I	US	2019-05-06	44 Min	F2F
R2	3	Chief Technology Officer	Parent Company Alfa	SWE	2019-05-13	31 Min	VIDEO

which guided the inductive theory development in subsequent steps.

3.3. Data analysis

We used the Gioia analysis, which is a structured yet flexible approach to analyzing qualitative data (Gioia et al., 2013). Our data analysis was carried out in three steps:

Step 1. Develop first-order categories and tentative second-order themes

The formal analysis began as we immersed ourselves in the data by reading it repeatedly and independently, in order to get an understanding of its depth and width. To ensure that there was a logical link between the interview questions and first-order statements, we paid extra attention to the initial interview protocol and the research questions. The first reading was made without taking notes, to ensure that patterns were only shaped after taking all the data into account. After the first reading, the analysis continued with open coding, in which statements related to each research question were highlighted. All statements were collected and first sorted on their relevance to each research question, and then sorted on their connection to either NIH or NSH. The procedure of axial coding followed, where related codes were connected to broader yet preliminary second-order themes. For example, statements coded as “fear of losing control of technology development”, “only accept technology according to established standards”, “only consider knowledge from a trusted colleague” and “excessive guidelines to avoid unwarranted IP” were categorized as “Obsessive control” whereas statements about “anxiety of not living up to expectations”, “procedural knowledge is not seen as competitive” and “domain-specific knowledge is seen as the main asset” were categorized as “Depreciation of knowledge”. To ensure that we adhered transparently and faithfully to informant terms, we made little attempt to refine the first order codes. This resulted in a rather large set of first-order codes and tentative second-order themes of causes, consequences and mitigating mechanisms.

Step 2. Iterate between coding scheme, data and the literature

The second step started with an iteration of the first-order codes and tentative second-order themes, to further refine and improve accuracy. In this step, we tried to figure out the larger narrative of the collected data—answering the important question “What’s going on here?” by narrowing down the link between first-order codes and second-order themes. This step included extensive discussions regarding the relevance of the second-order themes, revisiting the data and reviewing the literature for further refinement. When refining the second-order themes, we paid attention to their reflection of the first-order codes. For instance, first-order codes related to an individual’s difficulty in understanding confidential knowledge were first organized under a second-order theme labeled “Protection of unique knowledge”. This, however, was later relabeled to capture the essence of the root-cause, resulting in the theme “Confidentiality unawareness”, i.e. capturing the difficulty in distinguishing between confidential and public knowledge. This step further involved a revision of the first-order codes to ensure that they formed a coherent pattern from more than one respondent.

Step 3. Generate aggregated theoretical dimensions

In the third step, we created aggregated theoretical dimensions to delineate how the second-order themes were related. This implied a continuous iteration by referring back to the dataset to check whether they were suitable or if any of the first-order codes, second-order themes or aggregated theoretical dimensions needed further clarifications and/or amendments. The evaluation of possible models demonstrated that the causes of NIH problems were best illustrated by a map through the aggregated dimension of *Desire for excellence*, and NSH problems by *Fear of giving without receiving*. The consequences of NIH problems were best illustrated by a map through the aggregated dimension of *Resource inefficiency*, and NSH problems by *Painting oneself into a corner*. The

mitigating mechanisms of NIH problems were best illustrated by a map through the dimension of *Alignment of objectives*, and mitigating mechanisms of NSH problems by *Creation of a knowledge sharing environment*. The last step consisted of analyzing the second-order themes in relation to the purpose of the study in a framework, connecting causes and consequences with the mitigating mechanisms.

As a last step, we went back to evaluate all the data and merge our results into a conceptual framework, thus strengthening the contributions of the inductive study by generating testable propositions (Gioia et al., 2013). The connections in the framework emerged as a result of analyzing how causes, consequences and mechanisms were associated and mentioned in the data, while the strength of a connection was defined by the frequency with which an association was repeated in the data. All this work resulted in an initial framework. This initial framework was then tested in a workshop in which six interview respondents from three different CIHs were assigned to draw connections and their prevalence among the causes, consequences and mechanisms, without knowing about the initial framework. The respondents arrived at similar connections with minor differences, such as to the connections between the mitigating mechanisms of NIH problems and their causes and consequences. Half of the respondents only perceived the mitigating mechanism “translate relevance of ideas” as having a strong effect on the consequence of lagging behind the knowledge frontier, whereas the others also perceived it as having a strong effect on the suffocation of external ideas. In addition, the ones who perceived it as having a strong effect on the suffocation of external ideas also drew lines to both of the mitigating mechanisms for NSH problems, arguing that both are needed in order to mitigate the causes of NSH problems. The outcomes of this exercise were later taken into consideration before finalizing our framework.

4. Results and analysis

This section presents the findings from the inductive analysis; see Fig. 1 for the data structure. Fig. 1 shows six dimensions representing the causes, consequences and mitigating mechanisms of the NIH and NSH problems that arise when collaborating through a CIH. Tables 3–5 provide representative quotations to support Fig. 1.

4.1. Causes of NIH & NSH problems arising when collaborating through CIHs

4.1.1. Causes of NIH problems

4.1.1.1. Desire for excellence. This aggregate dimension highlights the drive within the studied organizations collaborating through a CIH to be at the front of the race. When collaborating through a CIH, startups argued arrogantly that their main asset is their knowledge, and at the same time, parent companies were sometimes equally arrogant, stating that the superiority of their knowledge could not be questioned. Three second-order themes were identified for the causes of NIH: *Exaggerated knowledge valuation*, *Obsessive control* and *Internal antagonism*.

Exaggerated knowledge valuation. This second-order theme focuses on the generous self-evaluation of knowledge assets among organizations, especially concerning domain-specific technical knowledge. All respondents perceived others’ knowledge to be of less value than their own and they also expressed difficulties in absorbing knowledge that was similar to their own. As the CEO and Co-founder of startup Iota (R29) stated:

“I don’t think that we learned anything from the domain expertise when collaborating through the CIH, we still have the upper hand when it comes to that.”

Furthermore, all respondents expressed a fear of not being needed and of not being able to contribute to their organizations. This fear

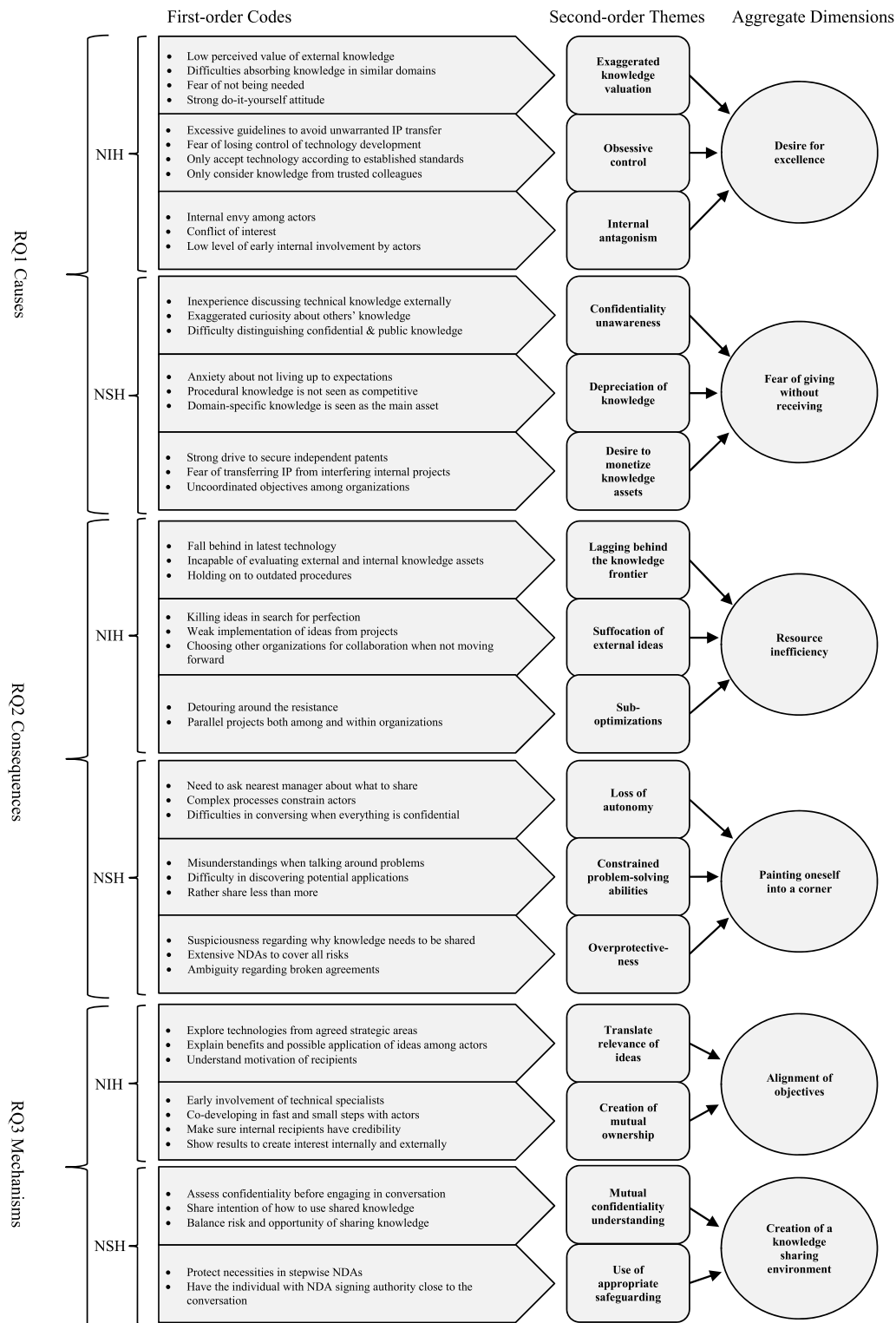


Fig. 1. Data structure.

might be linked to the strong do-it-yourself attitude, where organizations become defensive when external organizations question and challenge their knowledge, creating NIH problems among organizations when collaborating through a CIH.

Obsessive control. This second-order theme centers on the fear of losing control of one's technology development, where organizations

expressed difficulties in adapting external knowledge due to the fear of becoming too strongly influenced by others and dependent on them. Respondents said that parent companies only accepted technology according to established standards, whereby external technology must be controlled to ensure it fits within their rigorous specifications. Moreover, knowledge is only perceived to be worth absorbing if it comes from

Table 3
Representative supportive data for RQ1, causes of NIH & NSH problems.

Dimension & themes	Representative quotes (CIH: Corporate Innovation Hub, ST: Startup, PC: Parent Company)
CAUSES OF NIH	
<i>Desire for excellence</i>	
<i>Exaggerated knowledge valuation</i>	“How hard can it be is what they say? How hard can it be? They’ve only put five years of effort into this startup so far. And what they don’t count is the domain expertise you brought to it.” (R14, ST) “There are engineers undermining the difficulties, you meet a startup that’s been in existence for 10 years that solved a technical problem and then the engineers say, how hard can it be? In our parent company, we have many engineers, we must be able to solve this too.” (R10, CIH) “Often, there’s a bit of a disconnect, let’s say, a brilliant smart guy in machine learning from a startup and then you have an electrical engineer, mechanical engineer, let’s say at a known OEM. They can talk past each other. Not because they’re bad people or intentionally trying to trick each other but because they just approach the world in different ways. I’ve seen many smart startup engineer’s technologists that would say: That’s silly, that’s so completely the wrong way to think about it, we’re just going to replace all that and it’s going to be solved in this way.” (R30, ST)
<i>Obsessive control</i>	“The second one is losing control of your technology yeah and losing sight of what your vision is as a company, and the third one is losing your competitive edge and being unneeded in industry.” (R29, ST) “We realized quite quickly that we will not trust 100% of what the startup says, we will have to follow up that they have done solid safety checks.” (R27, PC)
<i>Internal antagonism</i>	“Then you feel some envy, why are they in Silicon Valley allowed to sit there and come up with a lot of cool stuff? I’m only allowed to stare at quality problems.” (R10, CIH) “Was extremely difficult to get it through, because they at the parent company weren’t taking part in the concept development, they lacked context and could not understand the specific design choices we made and therefore didn’t buy it.” (R12, CIH)
CAUSES OF NSH	
<i>Fear of giving without receiving</i>	
<i>Confidentiality unawareness</i>	“Of my knowledge I might have 25% that I can share and 25% that I know that I cannot share. 50% is quite unknown and in most of the cases within this part, I turn to my supervisor to find out if it’s allowed to be shared or not.” (R27, PC) “Right so there’s a bit of a give and take but you often get into situations where there is no give they just constantly want more, more, and more. You sit there like you’re being interrogated right, I’m exaggerating a little bit, but I think that’s the biggest concern, it’s the fundamental IP and protecting the IP.” (R30, ST)
<i>Depreciation of knowledge</i>	“If startups see what data we have collected from our vehicles and they realize they it can be replicated through a simple cell phone, then it becomes quite tough for us. So, what’s the fear? It’s that they might realize that all we have might not be that valuable as we believe.” (R23, CIH) “I believe that domain-specific knowledge is more sensitive, it’s what’s building the value and the uniqueness of a company, rather than the process of how it is performed, e.g. a development project or a POC, or a manufacturing process, the uniqueness of a company is broken down into domain knowledge.” (R5, ST)
<i>Desire to monetize knowledge assets</i>	“It’s actually a huge challenge to find data. We can share data with startups to do a POC, proof of concept. But that’s where you’d get into GDPR regulations and start to question who really owns the data? Is it ours since it was collected by us? But if it’s collected by us and another company, then it’s a tug-of-war on who owns the data, how we can utilize that and do what. I think that’s kind of it’s one of the things where it’s really slowing some things down. So right now, we actually have a project of just trying to produce some data for POC’s only.” (R16, CIH) “We have our own technology project in this domain and we had even before the startup was established so we wanted to keep our own technology development within the parent company quite separately from collaboration ... because we wanted to minimize the risk of that IP from us would slip into them.” (R27, PC)

someone you trust, preferably your nearest colleague at the company. As a Tech Scout at CIH IV (R24) explained:

“So, when we do corporate research, we would never advertise it until it is kind of perfect, especially in automotive we have a lot of guidelines and it has to fulfill some specific features.”

Internal antagonism. This second-order theme centers on the perceived conflict of interest internally among parent companies and CIHs. The CIH and their parent company express internal envy towards each other due to limited understanding of each other’s work, where both sides behave arrogantly, perceiving themselves as more valuable than the other. This envy frequently leads to a lack of motivation among individuals in organizations to onboard knowledge from each other and also to organizations starting to question what they will gain from it. As a Business Developer at parent company Beta (R25) said:

“They have to start to communicate and not be the secret army that’s simply sitting in Silicon Valley and does their stuff and believes they can do it so much better.”

Besides, the parent companies rarely managed to become involved with the CIH-led collaborations at an early stage, and this led to a lack of understanding and limited contextual insight into the CIH. It is therefore difficult for them to accept new knowledge from the CIH.

4.1.2. Causes of NSH problems

4.1.2.1. Fear of giving without receiving. When collaborating through a CIH, organizations are constantly engaging in a process of reciprocal

knowledge trading, so this dimension centers on each organization’s desire to profit from the most valuable knowledge assets, while less desirable knowledge is overlooked. This dimension has three themes: *Confidentiality unawareness*, *Depreciation of knowledge* and *Desire to monetize knowledge assets*.

Confidentiality unawareness. This second-order theme demonstrates the low awareness of what is confidential as opposed to public knowledge that is shown by individuals in organizations collaborating through a CIH. When they are not clear as to what is confidential, individuals tend to ask inappropriate questions about others’ knowledge. It often seems as though individuals in organizations are inexperienced in discussing advanced technologies externally, and are largely unaware that research has often become public knowledge. This creates unnecessarily anxious and restrictive behavior in discussions. A Venture Capitalist at CIH IV (R22) expressed awareness of confidentiality in the following way:

“Experts who are working at the CIH, when they talk to an external party for the first time they are really afraid of talking to them and sharing anything, because if you are not used to it, you just don’t know, you think everything is confidential, but it is actually not.”

This implies that knowledge will most likely not be shared by employees of companies whose legal representatives state it is up to each individual to assess confidentiality but individuals claim that, on the contrary, it is the legal entities’ responsibility.

Depreciation of knowledge. This second-order theme highlights each organization’s insecurity about valuing knowledge, especially procedural knowledge, where individuals in organizations express

Table 4

Representative supportive data for RQ2, consequences of NIH & NSH problems.

Dimensions & Themes	Representative quotes (CIH: Corporate Innovation Hub, ST: Startup, PC: Parent Company)
CONSEQUENCES OF NIH	
Resource inefficiency	
<i>Lagging behind the knowledge frontier</i>	“Their reaction is that this is not at all a good idea, what we do here at the parent company is much better ... And it is not so easy to compare this, because there is not a scale for comparing how good something is.” (R26, PC) “My guess is that you can’t use the same working methods anymore, so sometimes I wonder if the parent company’s people are stuck in their procedural knowledge, and that it prevents them from developing anything.” (R23, CIH)
<i>Suffocation of external ideas</i>	“You acquire the technology and the people, but you don’t implement it properly and the whole thing dies and then you are there with people you don’t need anymore, or they are unhappy because they see their ideas going down the drain and they are not producing anymore because they are unmotivated, and you waste money for nothing.” (R22, CIH) “We get stuck in multiple processes, but we just want to discuss and solve the client company’s problem, we want to be able to sell and don’t want to be treated as if we were on some sort of speed-dating. We want to sell, we are a company and we run a business and just because we’re a startup we don’t want to be a part of someone’s innovation statistics” (R5, ST)
<i>Sub-optimizations</i>	“We’re not replacing what you do, you’re not supposed to stop what you’re doing at the parent company, but just in case this doesn’t work out, let’s have a separate project that is parallel to this.” (R30, ST) “We had been able to bring in people from the parent company who have worked within this field for a very long time, where the startup had kind of zero knowledge. So, I think we could have cut huge amount of time if we had brought our resources to the project.” (R27, PC)
CONSEQUENCES OF NSH	
Painting oneself into a corner	
<i>Loss of autonomy</i>	“Lots of times in this technology project I did not make the decision myself to share or not share, I asked my supervisor.” (R27, PC) “I believe there is a risk of closing the relationship if you say that we will talk only about this, only this is confidential, then you might create an even greater fear of having a free discussion in these projects.” (R31, PC)
<i>Constrained problem-solving abilities</i>	“It turns into a catch 22 because you want to know what they can help us with, but we cannot tell them that. It’s definitely a problem and it was difficult to understand how you were supposed to cope with it.” (R12, CIH) “And so that’s the hardest thing for us to have the time to know about other large companies like company is where does company want to be in 10 years? It allows us to make sure that problems that we’re working on are the big issues to big companies and that we can be part of a solution to help that and we might be able to accelerate the that solution.” (R14, ST)
<i>Over-protectiveness</i>	“If you forget to empty the dishwasher once, I will ask for a divorce, and I believe that’s the big why people don’t share knowledge since there’s a huge ambiguity in the relationship.” (R10, CIH) “And then two months later, when it comes to contract, she pulls out this contract or hands you to a legal person and the contract basically says you’re selling your company to us for this project. The trust is broken.” (R30, ST)

anxiousness about not living up to the other organization’s expectations. Procedural knowledge is most often overlooked, since it is not considered as competitive in comparison to domain-specific technical knowledge. As a result, it is either shared without anyone noticing or considering its importance, or not at all. As the Chief Technology Officer at parent company Alfa (R2) said:

“There’s no one in the world that wants to share processes, no one wants to listen to it either. You will never be able to outrun anyone by copying someone else’s working methods.”

Desire to monetize knowledge assets. This second-order theme demonstrates each organization’s fear of not being credited for their knowledge. Knowledge that can be translated into monetary terms is tightly protected to ensure maximum value in return. The tug-of-war regarding IP among organizations collaborating through CIHs is sensitive. For instance, parent companies keep their specialists in certain areas away from projects that are operating through the CIH in their domain, to minimize IP being leaked to startups. At the same time, startups say that IP-related knowledge is their only asset, and that it will not be shared until they are confident it will stay with them. The CEO and co-founder of startup Pi (R30) said:

“At the end of the day, the reason why a small tech company can grow and continue to attract capital and relationships and do business is only because of specific, unique knowledge ... So the only way to monetize that or to prepare is to protect that.”

Additionally, in most cases, both organizations were afraid of ending up in legal disputes over exactly what knowledge each individual in each respective organization possessed before the project.

4.2. Consequences of NIH & NSH problems arising from collaboration through CIHs

4.2.1. Consequences of NIH problems

4.2.1.1. Resource inefficiency. This dimension centers on the ineffective use of resources when collaborating through a CIH, because it is difficult for individuals in organizations to remain updated when discussions and decisions are delayed and drag on due to high NIH tendencies. Three second-order themes were identified for the consequences of NIH: *Lagging behind the knowledge frontier*, *Suffocation of external ideas* and *Sub-optimization*.

Lagging behind the knowledge frontier. The first second-order theme highlights that NIH problems generate difficulties in assessing external knowledge and keeping up with the latest technology, since organizations are ignorant of existing knowledge gaps. The strong belief in own knowledge hinders the search for new technologies and makes it near-impossible to evaluate internal and external knowledge assets when collaborating through a CIH. As expressed by the CEO and co-founder of startup Pi (R30):

“Over the long term, it can hollow out the company, when work continues to go on without anyone realizing the company is slowly falling behind the latest technology.”

Suffocation of external ideas. In the search for perfection, CIHs are torn between startups and the parent company when trying to keep ongoing collaborations alive. When these collaborations terminate, this will most often be due to lengthy processes and demanding standards. Resources invested, for example in the development aimed at obtaining

Table 5
Representative supportive data for RQ3, mitigating mechanisms of NIH & NSH problems.

Dimension & themes	Representative quotes (CIH: Corporate Innovation Hub, ST: Startup, PC: Parent Company)
MECHANISMS TO MITIGATE NIH	
<i>Alignment of objectives</i>	
<i>Translate relevance of ideas</i>	“I think what matters is having the connection and ability to go to Sweden and to figure out how we at the CIH fits into the mission of the parent company as a whole. Once the CIH and the parent company understand each other’s situation people at the parent company might think differently, the barrier of the not invented here starts to drop. Maybe the idea didn’t originate from the parent company, but the CIH don’t have the resources to do it, but the parent company does, and they can help the CIH. On the flip side of that, a startup that comes with an idea to the CIH, they have to be willing to let that go of it to some extent, like we have to be willing to let it go of ideas to the parent company” (R28, CIH) “Basically, proceed from a pointed-out technology stack where we have already realized that we need to partner up with others, and if something lies within that stack it would be easier.” (R27, PC)
<i>Creation of mutual ownership</i>	“They must have expertise in place at the CIH, because otherwise the things that are getting done will not be relevant, we just did a POC within AI with a startup where we sent a specialist for two weeks, and they worked from morning to evening, together with the startup and then went home, it was successful.” (R26, PC) “The CIH have to involve right people from the beginning, the faster you do that the faster people become receptive, people always seem to be more reluctant when handed a project too late in the process. You need to get people at the parent company to feel co-ownership and that their ideas are put into a context, then it is really difficult for them to say, no, we don’t want to do this, they have to feel co-ownership early.” (R5, ST) “Go out and present what the CIH are doing, how they work, ask if they want to be involved, show the benefits. You have to do this over and over, now I feel that it starts to go where we want.” (R25, PC)
MECHANISMS TO MITIGATE NSH	
<i>Creation of a knowledge sharing environment</i>	
<i>Mutual confidentiality understanding</i>	“Persons has to go through some form of car-wash on the road to the CIH, we have to make sure that the culture here is cared for, that it is open-minded and forward-leaning. If you fly in someone we have to tell them that all NDA’s are signed, all internal patents are managed, the startup’s patents are managed, during this period you are colleagues, your job is to ensure that this POC succeeds.” (R10, CIH) “Just being completely honest and upfront with the parent company, helps a lot to like, this is exactly what I intend to use it for. For this reason and if you’re okay with that then I hope that you’ll send that in and if that person is not okay with it then let’s have a discussion as to why you’re not okay with it.” (R28, CIH) “Early on we came up with an interface project, an early interface project very clearly defines the role of each partner and the defining what knowledge is needed to be shared among each actor needed for that role so it, this party is developing the engine controller and that’s it, therefore the knowledge share needs to be enough for them to share to come up with that.” (R21, ST)
<i>Use of appropriate safeguarding</i>	“Have special contracts in place for startups. It was quick decision making, they weren’t trying to get everything, all the IP and things like that. It’s was very flexible and reasonable for what we’re trying to do.” (R30, ST) “Instead of it being so very formal when signing an agreement, it should be seen as a natural part of a work ... when you step into the CIH you write a confidentiality agreement, on a digital tablet, with a preface form, that people should be able to sign in 2 min, just tick a box and when you start to develop something, you write a collaboration agreement that is fairly standard.” (R31, PC) “Coming from a previous organization, the data was made available to any group and it didn’t really matter whether you were doing engine design or engine development, whether you’re doing electrical control systems for the engine, just everyone had access. Even the engine builders had access to it. I think that because of this expectancy of data availability, being in a data enriched environment help people make better decisions.” (R28, CIH)

a patent, or in ongoing partner relationships, will be wasted when startups are acquired by others or when they choose to collaborate with other organizations. This has negative implications for the reputation of the CIH and its associated parent company. As a Venture Capitalist at parent company Alfa (R7) stated:

“This resulted in a situation where we became restrictive and suddenly leaned towards the parent company and told the startup to continue the collaboration on their own, and every time they presented a new solution, much time was spent on finding errors and telling them to come back when it is perfect.”

Sub-optimizations. This second-order theme highlights the consequences of the actions that are taken today towards solving NIH problems among organizations. According to startups and CIHs, you have to find ways to override the resistance in the parent company by, for example, going up hierarchies to receive knowledge or expert advice. As the Director of CIH III (R16) explained:

“You basically have to find ways around that person at the parent company. With some people you just cannot win.”

This will further affect implementations when people who are supposed to be involved within specific projects through the CIH are bypassed, creating even more resistance. In addition, sub-optimizations occur, such as when there are parallel projects, i.e. overlapping projects among organizations that run separately instead of being aligned.

4.2.2. Consequences of NSH problems

4.2.2.1. Painting oneself into a corner. This dimension emphasizes the fact that, when collaborating organizations cannot solve the problem they set up, they then become restrictive, resulting in a situation in which they do not know enough about the other party’s knowledge. Three second-order themes were identified for the consequences of NSH: *Loss of autonomy*, *Constrained problem-solving abilities* and *Over-protectiveness*.

Loss of autonomy. Due to the low level of awareness among individuals in organizations as to what is confidential knowledge, employees frequently ask their immediate manager about what they are allowed to share. However, if the manager does not have the answer, organizations collaborating through a CIH will become constrained as to how to move forward in projects on their own. This consequence might be particularly problematic for CIHs connected to parent companies in other time-zones, since it increases the likelihood of delaying dialogues. A Senior Software Developer at CIH II (R28) describes it in the following way:

“We have to adapt our schedules in order to help the relationship to grow with the parent company and develop it into a point where people over there feel comfortable actually providing us with knowledge that we need in order to get something done.”

Constrained problem-solving abilities. When organizations that

collaborate through a CIH share less rather than more, and talk around problems instead of talking directly about them, it will lead to misunderstandings and negatively affect the quality of work. Likewise, when employees at different hierarchical levels engage in discussions, they will most likely talk past each other. These misunderstandings will not only hinder the discovery of potential applications for new ideas but also increase the difficulty of assessing how well a solution fits into another company's product portfolio, making it almost impossible, as expressed by a Tech Scout at CIH IV (R24):

"You always have to kind of talk around things, you cannot be specific, if you could be specific you could say okay we are doing this and that and we want to have that it would make things sometimes a bit easier."

Over-protectiveness. To ensure a company will be able to monetize its own knowledge assets, extensive protection and NDAs are often used. Organizations face a wall of suspicious employees, and this makes it difficult to progress the collaborations through a CIH. A Senior Software Developer at CIH II (R28) describes a situation when trying to get knowledge from the parent company:

"You've seen as an outsider when working at the CIH and you've kind of shaking things up and asking questions that make people at the parent company feel uncomfortable. It's like, I need access to some data to do something. And it's like, why do you need access? It's like, Well, why do I even have to ask for access? It should be readily available."

4.3. Mechanisms mitigating NIH & NSH problems through CIHs

4.3.1. Mechanisms mitigating NIH problems

4.3.1.1. Alignment of objectives. This dimension emphasizes aligning objectives among organizations engaging in collaborations through a CIH in order to get them all on the same page when transferring knowledge. Two second-order themes were identified as mitigating mechanisms of NIH problems: *Translate relevance of ideas* and *Creation of mutual ownership*.

Translate relevance of ideas. To mitigate NIH, it will be important for the CIH to understand their position vis-a-vis the parent company. This will provide employees at the CIH with the confidence to understand what they can do in these collaborations as well as the ability to explain the benefits of ideas from multiple perspectives. As a Venture Capitalist at CIH IV (R22) highlighted:

"Sometimes it's explaining the idea once more from our perspective, to explain that, no this is not what you tried, this is something else because of bla, bla, bla. Or talking to our people at the parent company and to figure out their motivations, I mean to some extent, because this takes time and we are not paid for that, but to figure out why the people are so reluctant."

For instance, ideas and new technology explored through CIHs would benefit, some respondents indicated, from being a part of a pre-agreed technology area of the parent company, aligned with the long-term strategy and supported by top management. These technology areas have to be communicated throughout the parent company, where we also believe it is important that the CIHs are clear to startups about which areas the parent company will pursue with external organizations through the CIH and which areas will be developed solely in-house. However, by CIHs translating the relevancy of ideas among organizations, NIH can be mitigated, since decisions will be mutually agreed, which makes knowledge transfer possible.

Creation of mutual ownership. Ownership agreements about projects, data-sets, technology advances, possible outcomes that might lead to patents—all these need to be established among organizations

collaborating through a CIH. These agreements will mitigate the tendency to NIH that can be encountered when technical specialists are involved early on in tech-scouting activities. A Chief Technology Officer at parent company Gamma (R26) explained the benefits of involving recipients early on, but also the importance of moving fast by taking small steps when collaborating:

"What seems to work is to involve the recipients from the parent company early in the process, and you make the project quite quick, you take small steps together with the startup, you try to get to a conclusion in 2–3 weeks, that's what we try to have as an aim."

In addition, the CIH needs to ensure that the recipients at the parent company have credibility, by using internal champions. These are trusted specialists within the parent company who can advocate for the CIH. Furthermore, it is important and necessary to communicate the results and knowledge learned from projects and collaborations through the CIH, not only internally to the parent company but also externally to startups and other organizations, in order to create the motivation to participate.

4.3.2. Mechanisms mitigating NSH problems

4.3.2.1. Creation of a knowledge sharing environment. This dimension centers on the creation of an environment where individuals in organizations collaborating through a CIH feel safe to share knowledge with each other. Two second-order themes were identified for the mitigating mechanisms of NSH: *Mutual confidentiality understanding* and *Use of appropriate safeguarding*.

Mutual confidentiality understanding. Being able to understand the difference between confidential and non-confidential knowledge is fundamental to mitigating NSH, where mutual includes all three organizations: the CIH, parent company and startup. When individuals in organizations do not know what they can share, they most often do not share at all. All organizations should consider assessing confidentiality before engaging in conversation. The CIH should encourage this assessment, as it will lead to a safer knowledge sharing environment. As the CEO and Co-Founder of startup Iota (R29) stated:

"Think strategically about what knowledge you can leave with others and what you should keep for yourself, having that understanding before you approach these problems is the key to success."

We believe this is especially important when inviting technical specialists to engage in a POC, so all organizations and individuals know what knowledge will be shared and why. Lastly, when the CIH encourages other organizations to share its intentions on how to use shared knowledge, this will build trust among collaborating organizations.

Use of appropriate safeguarding. Organizations should be encouraged by the CIH to ensure that individual IP is secure before they engage in collaborations, to avoid undesired legal twists. Furthermore, the NDAs used for collaborations through the CIH need to be reworked and simplified and should preferably have a stepwise format, so that they only provide the protection required at each step of the collaboration. In addition, the NDA signing process at the CIH should be less formal and a natural part of the working processes, so that decision-makers, such as legal advisors, are easy to reach if specific paragraphs in the NDA need to be adjusted. Also, respondents stated that it is important to have designated employees at CIHs that have the authority to sign the NDAs without further approval. As a Business Developer at parent company Beta (R25) explained:

"Make it easier in terms of who signs NDAs, it doesn't have to be some top manager, we have two designated people in each unit, and make it okay to sign them a little faster. Then find out what type of agreement you need. You can have a process map, if you want to do

this, and this is what you want to achieve, then you will need this agreement.”

However, appropriate contracts will not solely build a safe knowledge sharing environment—having knowledge available from both the parent company and the startup is fundamental to pursuing collaborations among organizations through a CIH.

4.4. Towards a framework for mitigating NIH & NSH problems

Based on the inductive analysis, a framework illustrating how CIHs can work to mitigate NIH and NSH problems in knowledge transfer has been devised and is presented in Fig. 2. The framework is formed by deploying the themes and dimensions from the empirical analysis as building blocks, and is thus supported by the data structure presented in Fig. 1. When reading from left to right, the connections between causes and consequences are visualized by horizontal arrows, either as strong in black or as weak in grey. Mechanisms are shown as moderators, which influence the strength of the relationship between causes and consequences (i.e. they mitigate or weaken the consequences). These are illustrated with vertical arrows in the same color scheme. The framework is a visual illustration of the findings, and provides a contingency perspective on how to match mitigating mechanisms with causes and consequences.

Among the causes of NIH problems, *Obsessive control* and *Internal antagonism* lead to the most consequences, namely, *Lagging behind the knowledge frontier* and *Suffocation of external ideas*, thus indicating that these two causes and consequences may be the most critical. *Translate relevance of ideas* seems linked to the most connections between NIH causes and consequences, making it the most important mitigating mechanism to consider. That said, *Creation of mutual ownership* may also be a powerful mitigating mechanism, while *Exaggerated knowledge valuation* is another key cause of NIH problems, leading to *Lagging behind the knowledge frontier* (among other things).

On the other hand, *Confidentiality unawareness* is a cause of NSH problems that has strong connections to all key consequences, indicating that it may be the most critical cause. However, the consequence of NSH

Constrained problem-solving abilities has strong connections to all causes, making it a critical consequence. Lastly, *Mutual confidentiality understanding* is the most important mitigating mechanism to consider, since it has the strongest connections between NSH causes and consequences. However, the *Use of appropriate safeguarding* can be a powerful mitigating mechanism, and *Depreciation of knowledge* and the *Desire to monetize knowledge assets* are key causes to consider.

However, the mitigating mechanisms of both NIH and NSH problems should also be considered simultaneously, since NIH and NSH constitute interconnected problems as CIHs take part in increasingly open innovation processes. These processes are characterized by the active transfer of technology both in and out through company boundaries. Therefore, the act of transferring knowledge among parent companies and startups through a CIH requires an explicit discussion about both NIH and NSH, since it is essential to mitigate both these problems for a CIH to operate effectively.

5. Discussion and implications

This article extends prior research on NIH and NSH problems in knowledge transfer occurring among organizations collaborating through a CIH (Berger and Brem, 2016; Fecher et al., 2020). As more and more large corporations create CIHs to facilitate knowledge transfer to and from partners, additional insights about organizing and managing CIHs are needed. Specifically, CIHs are created as bridges into innovative ecosystems (Walrave et al., 2018), yet often fail to meet the objectives set for them. NIH and NSH problems are at the core of such failures. The existing literature has identified several mitigating mechanisms for NIH and NSH problems in more general knowledge transfer settings, but has so far not provided specific insights into how they play out in the context of CIHs (Aquilani et al., 2017). Our study helps close this gap by studying the causes, consequences and mitigating mechanisms of both NIH and NSH. The study is particularly relevant as more CIHs are established, and as more firms move towards increasingly open innovation processes involving the active transfer of technology across firm boundaries. Moreover, Monteiro and Birkinshaw (2017) have investigated knowledge transfer between CIHs and parent companies but have

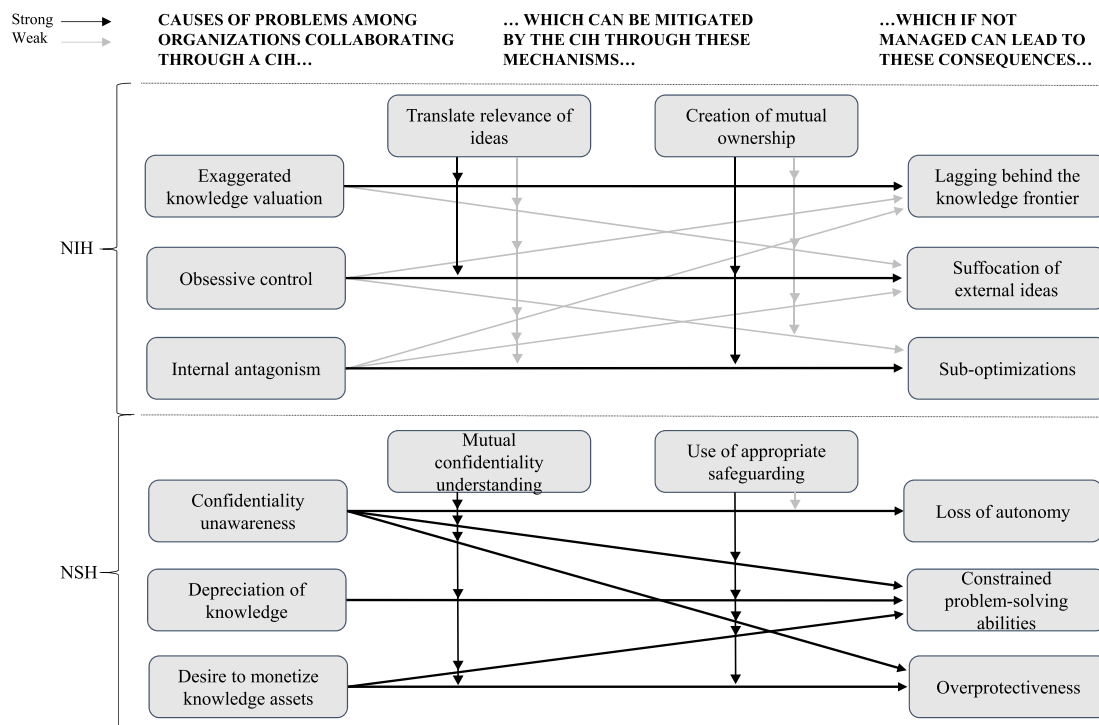


Fig. 2. CIHs mitigating NIH & NSH problems.

not provided insights into the knowledge transfer among CIHs, parent companies and startups. This increased interest in active technology transfer brings the problem of negative attitudes to the forefront (Aquilani et al., 2017; Chesbrough, 2003; De Araújo Burcharth et al., 2014).

5.1. Theoretical implications

In response to the sparse and largely conceptual research on NIH and NSH problems in knowledge transfer among organizations collaborating through a CIH, our article contributes to prior literature in the following ways.

First, we contribute to the conceptual understanding of NIH and NSH problems in the context of CIHs involving multiple organizations in an ecosystem setting (Adner, 2017; Banc and Messegem, 2020). Prior literature has so far mainly focused on these two problems in a two-sided context, i.e. in dyads (Antons et al., 2017; Antons and Piller, 2015; De Araújo Burcharth et al., 2014; Hussinger and Wastyn, 2012; Kathoefer and Leker, 2012). By studying NIH and NSH problems in a context where multiple organizations collaborate through a CIH, we not only find new causes and consequences for NIH and NSH problems, but also new insights into the corresponding mitigating mechanisms for both problems. By doing so we answer the calls from Hannen et al. (2019) and Banc and Messegem (2020) for an ecosystem perspective when studying NIH in knowledge absorption processes. For instance, the cause of NIH that we have labeled *Internal antagonism* seems neglected in prior literature, and clearly has its roots in an ecosystem context. Internal antagonism could also be interpreted as a sign of internal and individual competitiveness within an organization. Just as Hussinger and Wastyn (2012) argued that NIH occurs when transferring knowledge between competitors, our findings emphasize that organizations collaborating through a CIH could sense a strong competitive situation even if they belong to the same organization. Competition is indeed a key issue in ecosystems (Granstrand and Holgersson, 2020).

Likewise, the cause of NSH that we labeled *Depreciation of knowledge*, which could be interpreted as a sign of fear of not being an attractive partner when collaborating with external organizations, seems to have been overlooked in prior literature. Organizations collaborating in ecosystems seem not only to be afraid of sharing too much, but also of sharing knowledge that might not be as appealing for others to absorb. This might be an effect of limited prior experience with outward technology transfer (Hoegl et al., 2011). Another finding that seems neglected in prior literature is the mitigating mechanism of NSH that we labeled *Mutual confidentiality understanding*. This underscores the importance of experience in knowledge transfer in an ecosystem context and aligns with Aquilani et al. (2017) in developing a strong sense of trust and openness when engaging in knowledge transfer in a CIH context. Mitigating NIH and NSH problems in ecosystem contexts may therefore be more complex than it has been depicted in the literature thus far. And in light of increasingly open innovation processes (Adner, 2017; Chesbrough, 2003), with multiple organizations involved, NIH and NSH problems need to be studied in ecosystems rather than just between two entities. Even though a CIH is established to create de-biasing bridges among collaborating organizations (Berger and Brem, 2016), the bridges among all organizations will still be guarded by the negative attitudes resulting from NIH and NSH biases. We thus propose that the NIH and NSH literature should better incorporate multi-actor collaboration involving intermediaries.

Second, our results shed new light on knowledge transfer in two-way reciprocal processes, where NIH occurs on one side of the process and NSH on the other (Chesbrough, 2003). In prior literature, NSH is largely overlooked and potentially underestimated (Antons and Piller, 2015; Hannen et al., 2019), although the process of sharing knowledge is heavily affected by the NSH problem in a CIH context (Hannen et al., 2019). The action of absorbing knowledge by one firm thus requires the fundamental action of sharing knowledge by another. For example, one

organization's NIH attitude, such as *Obsessive control*, could lead to another organization's NSH consequence (such as *Loss of autonomy*). While our study was not designed to perfectly match NIH and NSH problems, it underscores that individuals adopting boundary-spanning roles may act to identify and mitigate both NIH and NSH attitudes when transferring knowledge across organizational boundaries. Individuals at CIHs typically assume boundary-spanning roles (Monteiro and Birkinshaw, 2017; Tracey et al., 2018). To improve knowledge transfer in ecosystems as well as in dyads, we need to study NIH and NSH problems in more depth, and—most importantly—*simultaneously*. Our study is a step in that direction, but we call for future research to further advance the understanding of how these two problems interconnect in knowledge transfer.

Third, our research contributes by providing new causes, consequences and mitigating mechanisms for both NIH and NSH problems, and further delineates how certain consequences can be mitigated by specific mechanisms. Any organization involved in technology transfer needs to manage conflicts and/or deal with the resistance that follows from these problems (Calof et al., 2018). However, as prior literature has mainly provided, and adopted, one-fits-all mechanisms for both NIH and NSH problems, our findings provide more fine-grained knowledge of how to best mitigate NIH and NSH problems. For instance, Aquilani et al. (2017) and Longo and Giaccone (2017) argue that the mechanism to build a strong sense of trust and openness can mitigate both NIH and NSH problems at the same time. While these results make sense on an overall level, our research underscores that the key mitigating mechanisms for NIH problems may be different from those for NSH problems. In other words, even though the two problems are connected in open innovation processes (Chesbrough, 2003), they seem to have fundamentally different root causes and consequences, which need to be considered in choosing appropriate mitigation mechanisms. This finding sets out a contingency perspective for thinking about how to better mitigate NIH and NSH problems, and extends prior literature on CIH-related NIH and NSH (Aquilani et al., 2017; Longo and Giaccone, 2017).

5.2. Managerial implications

A corporate innovation hub constitutes a specific governance structure, where the hub largely decides who is allowed to participate and how participation takes place (Remneland Wikhamn and Styhre, 2019). The causes, consequences and mitigating mechanisms we have identified can thus be used to improve this governance structure by detecting, mitigating and/or preventing NIH and NSH problems. CIH managers and collaborating organizations—startups and parent corporations—can benefit from the framework, which provides a guide and a blueprint for proactively preventing tensions and misunderstandings among collaborating organizations. Specifically, the results have three different implications. First, awareness of the causes identified can assist all collaborating organizations to identify NIH and NSH problems in and among a group of organizations collaborating through a CIH. They also provide a tool for self-assessment and group discussion, e.g. on whether low confidentiality awareness is a problem, how that manifests itself, and what potentially could be done.

Second, CIH managers, in particular, can seek to apply mechanisms for specific causes they have experienced—bearing in mind that NIH and NSH problems need to be mitigated independently—where some mechanisms seem more appropriate than others, depending on the cause. For NIH, the mechanism *Translate relevance of ideas* may be suitable for addressing *Exaggerated knowledge valuation* and preventing the consequence *Lagging behind the knowledge frontier*. And for NSH, an example would be to use the mechanism *Mutual confidentiality understanding* when mitigating *Confidentiality unawareness* and preventing the consequence of *Constrained problem-solving abilities*. In addition, the causes and consequences can be used by managers at CIHs to ensure legitimacy when advocating for the importance of CIH activities,

including promoting the value proposition of such activities to internal stakeholders. By doing so, it can become more than just an “innovation theatre”.

Third, we believe our results will also be applicable to parent companies and startups. Specifically, these organizations can increase their awareness of how their behavior can affect collaborations through CIHs. We hope that, by becoming more aware, both types of organizations will seek to implement their own NIH and NSH problem-mitigating mechanisms when collaborating through a CIH.

5.3. Limitations and future research

This study comes with a couple of important limitations. First, all investigated CIHs are active in exploring technologies related to future transportation solutions. This reduces generalization of the results, which calls for future studies in other industries to understand the causes, consequences and mitigating mechanisms for NIH and NSH. It is important to keep in mind that the transportation industry is somewhat conservative about leveraging external innovation (Hjalmarsson Jordanius et al., 2019), which makes it a good case for studying NIH and NSH problems. Second, there are other types of organizations collaborating through a CIH in addition to those investigated in the current study (see Pauwels et al., 2016; Salomaa, 2018; Schuh et al., 2017). These include academia, investors and government, and this implies the need to further our understanding of NIH and NSH problems in a broader set of organizations collaborating through a CIH.

The inductive case study also has important limitations as a method, namely a limited sample and a heavy reliance on key informants. Additionally, even if the opinions, ideas and thoughts of participants in the innovation system (CIHs, parent companies, partners, etc.) are important, the real impact may differ from these opinions, ideas or thoughts (Mees-Buss et al., 2020). To offset these weaknesses, we triangulated the results, but we also encourage future studies into NIH and NSH problems in and around CIHs which will encompass a broader set of organizations, across other industries and technological domains, and with different research designs. Our research lays the basis for further quantitative research which can validate the relationship between the causes and consequences while testing the effectiveness of the mechanisms to mitigate NIH and NSH.

Moreover, we encourage more specific studies into the actions that different types of organizations can take to mitigate NIH and NSH problems. Different types of organizations may vary in agency, motivations and capabilities. Another topic ripe for future research is that of the interrelationships between NIH and NSH. In the current study, these were treated largely as independent problems, but in open innovation processes of a coupled type, which are characterized by the simultaneous transfer of technology in multiple directions, more research is needed into how NIH affects NSH and vice versa.

Next, our findings indicate that NIH and NSH problems may be more prominent when transferring domain-specific technical knowledge than procedural knowledge. This aligns with Schuh et al. (2017) and Antons and Piller (2015), who argue that the type of knowledge transferred impacts NIH problems. It would be useful to investigate whether this also holds for NSH problems, and whether the causes, consequences and mechanisms differ depending on which knowledge type is being transferred. In addition, future research should dive deeper into the concept of NSH and its constituent elements because causes, consequences and mitigating mechanisms may differ depending on whether the problem arises in the selling or sharing of knowledge or technology.

Finally, some CIHs seem to struggle with striking a balance between being independent from the parent company and not being disconnected. When striving for independence, the CIHs often become isolated without the support of the parent company, which in itself can generate NIH and NSH problems. Further study of this problem is particularly relevant, as an ever-increasing number of technology transactions go through CIHs.

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